

# Inference for Covariance Matrices

Readings: Johnson & Wichern (2007), Section 4.4;  
Rencher (2002), Sections 7.1–7.2

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# Agenda

Covariance Matrix

Statistical Inference for the Covariance Matrix

High-Dimensional Covariance Estimation

# Learning Objectives

By the end of this lecture, you should be able to:

- ▶ **Interpret and estimate a covariance matrix** and explain what its entries, eigenvalues, and determinant tell us about multivariate data
- ▶ **Understand sampling uncertainty in covariance estimation** and use the [Wishart distribution](#) as the multivariate analogue of the chi-square distribution for variance estimation
- ▶ **Perform basic inference for covariance matrices**, including confidence intervals and hypothesis tests
- ▶ **Recognize when the sample covariance matrix breaks down** in high-dimensional problems and explain how shrinkage and sparse estimation address these limitations

# Review: Interpretation of Covariance

- ▶  $\text{Cov}(X, Y) > 0$ : variables tend to increase together
- ▶  $\text{Cov}(X, Y) < 0$ : one increases as the other decreases
- ▶  $\text{Cov}(X, Y) = 0$ : no linear association

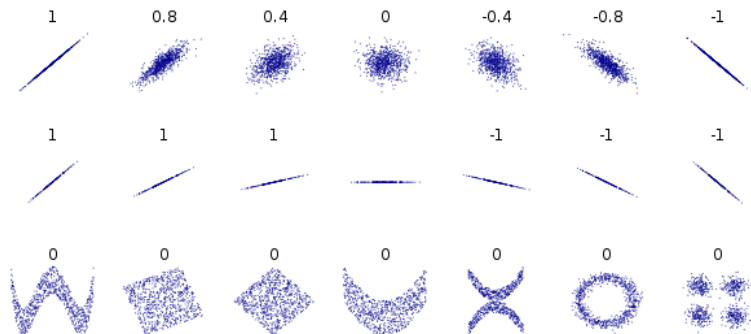


Figure 1: Image courtesy of Wikipedia at [https://en.wikipedia.org/wiki/Correlation\\_and\\_dependence](https://en.wikipedia.org/wiki/Correlation_and_dependence)

# Population Covariance Matrix

For a random vector

$$\mathbf{X} = (X_1, \dots, X_p)^\top, \quad E(\mathbf{X}) = \boldsymbol{\mu},$$

the population covariance matrix is

$$\boldsymbol{\Sigma} = E \left[ (\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top \right] = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1p} \\ \vdots & \ddots & \vdots \\ \sigma_{p1} & \cdots & \sigma_{pp} \end{bmatrix}.$$

- ▶ Diagonal:  $\sigma_{ii} = \text{Var}(X_i)$
- ▶ Off-diagonal:  $\sigma_{ij} = \text{Cov}(X_i, X_j)$
- ▶ Since  $\sigma_{ij} = \sigma_{ji}$ ,

$$\boldsymbol{\Sigma} = \boldsymbol{\Sigma}^\top.$$

$\boldsymbol{\Sigma}$  summarizes the **variability and linear dependence** among all variables simultaneously.

# Properties of Covariance Matrices

A valid covariance matrix must satisfy:

- **Symmetry:**

$$\Sigma = \Sigma^{\top}$$

- **Positive semi-definiteness:** for any vector  $\mathbf{a}$ ,

$$\mathbf{a}^{\top} \Sigma \mathbf{a} \geq 0,$$

because

$$\mathbf{a}^{\top} \Sigma \mathbf{a} = \text{Var}(\mathbf{a}^{\top} \mathbf{X}) \geq 0.$$

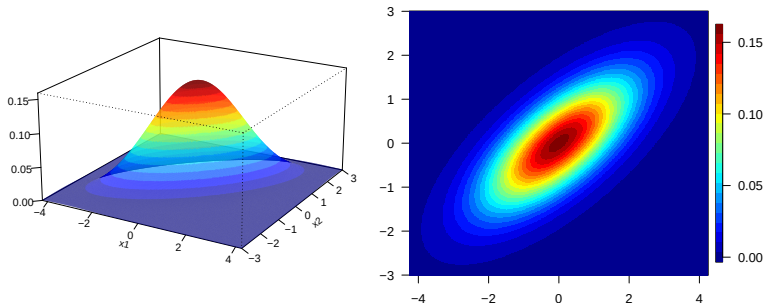
- **Nonnegative eigenvalues:** the eigenvalues describe variability along principal directions
- **Generalized variance:**  $|\Sigma|$  summarizes the overall multivariate spread

# Geometric Interpretation

- Covariance matrix defines **ellipses of constant Mahalanobis distance**:

$$D^2(\mathbf{x}) = (\mathbf{x} - \bar{\mathbf{x}})^T \Sigma^{-1} (\mathbf{x} - \bar{\mathbf{x}})$$

- Contours visualize variance and correlation structure



$\Sigma$  typically unknown and estimated using the sample covariance matrix

# Sample Covariance Matrix (SCM)

The population covariance matrix  $\Sigma$  is usually unknown.  
Given observations

$$\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p,$$

we estimate it using the **sample covariance matrix**

$$\hat{\Sigma} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top.$$

- ▶ Symmetric and positive semi-definite
- ▶ Estimates both individual variances and pairwise covariances
- ▶ Used throughout multivariate analysis: PCA, discriminant analysis, Mahalanobis distance, and many other methods

Key question: **How much does  $\hat{\Sigma}$  vary from sample to sample?**

## Review: Inference for a Univariate Variance

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2).$$

The sample variance is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2,$$

and

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2.$$

This sampling distribution allows us to construct confidence intervals and hypothesis tests for  $\sigma^2$ .

**Univariate:**  $\sigma^2 \longrightarrow s^2 \longrightarrow \chi^2$

**Multivariate:**  $\Sigma \longrightarrow \hat{\Sigma} \longrightarrow \text{Wishart}$

# Wishart Distribution: Sampling Distribution of the SCM

Suppose

$$\mathbf{X}_1, \dots, \mathbf{X}_n \stackrel{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}).$$

Define

$$\mathbf{S} = \sum_{i=1}^n (\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^\top.$$

Then  $\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$ , where  $W_p$  denotes the [Wishart distribution](#).

Since

$$\hat{\boldsymbol{\Sigma}} = \frac{\mathbf{S}}{n-1}, \quad E(\mathbf{S}) = (n-1)\boldsymbol{\Sigma},$$

we obtain

$$E(\hat{\boldsymbol{\Sigma}}) = \boldsymbol{\Sigma}.$$

The Wishart distribution describes [sampling variability of co-variance matrices](#), just as the chi-square distribution describes sampling variability of a univariate variance.

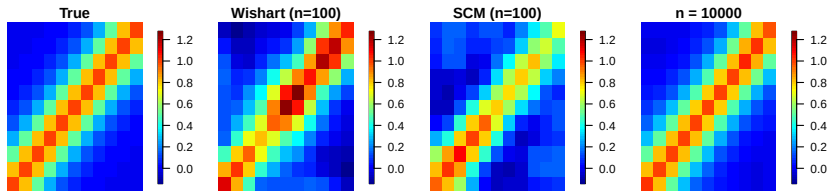
# Sampling Variability of the Sample Covariance Matrix

For  $\mathbf{S} \sim W_p(n-1, \mathbf{\Sigma})$ ,

$$\mathbb{E}[S_{ij}] = (n-1)\Sigma_{ij}, \quad \text{Var}(S_{ij}) = (n-1) (\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj}).$$

Therefore, for  $\hat{\mathbf{\Sigma}} = \mathbf{S}/(n-1)$ ,

$$\text{Var}(\hat{\Sigma}_{ij}) = \frac{\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj}}{n-1}$$



# Inference for Individual Covariance Elements

For a diagonal element  $\sigma_{ii}$ ,

$$\frac{(n-1)\hat{\sigma}_{ii}}{\sigma_{ii}} \sim \chi_{n-1}^2,$$

so we can construct an exact confidence interval for the variance  $\sigma_{ii}$  under multivariate normality.

For **off-diagonal covariances and correlations**, inference is less direct.

- ▶ Large-sample approximations can be used
- ▶ Bootstrap methods [\[Efron, 1979\]](#) provide a flexible alternative

**Bootstrap idea:**

resample observations  $\rightarrow \hat{\Sigma}^{*(1)}, \dots, \hat{\Sigma}^{*(B)} \rightarrow$  quantify uncertainty.

# Statistical Inference for $\Sigma$

Suppose we want to test whether the population covariance matrix equals a specified matrix:

$$H_0 : \Sigma = \Sigma_0 \quad \text{vs.} \quad H_1 : \Sigma \neq \Sigma_0.$$

- Under multivariate normality, the **likelihood ratio test (LRT)** statistic is

$$\text{LRT} = n \left[ \log |\Sigma_0| - \log |\hat{\Sigma}| + \text{tr} \left( \Sigma_0^{-1} \hat{\Sigma} \right) - p \right].$$

- Under  $H_0$ , for large  $n$ ,

$$\text{LRT} \sim \chi_{p(p+1)/2}^2,$$

where  $p(p+1)/2$  is the number of unique elements in  $\Sigma$ .

- For finite samples or more complex settings, **bootstrap methods** can be used for uncertainty quantification.

## Example: Hypothesis Test for $\Sigma$

Test  $H_0 : \Sigma = I$  with  $p = 2$ ,  $n = 50$ :

$$LRT = n \left( \log |I| - \log |\hat{\Sigma}| + \text{tr}(\hat{\Sigma}) - p \right)$$

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Suppose  $\hat{\Sigma} = \begin{bmatrix} 1.5 & 0.3 \\ 0.3 & 0.7 \end{bmatrix}$

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Compare with the  $\chi^2_3$  critical value of  $\approx 7.81$  at  $\alpha = 0.05$ . Since the test statistic (12.04) exceeds this critical value, we reject the null hypothesis that  $\Sigma = I$ .

# What Changes in High Dimensions?

The sample covariance matrix is

$$\hat{\Sigma} = \frac{1}{n-1}(\mathbf{X} - \bar{\mathbf{X}})^\top (\mathbf{X} - \bar{\mathbf{X}}),$$

Therefore,

$$\text{rank}(\hat{\Sigma}) \leq \min(p, n-1).$$

If

$$p \geq n,$$

then

$$\text{rank}(\hat{\Sigma}) \leq n-1 < p,$$

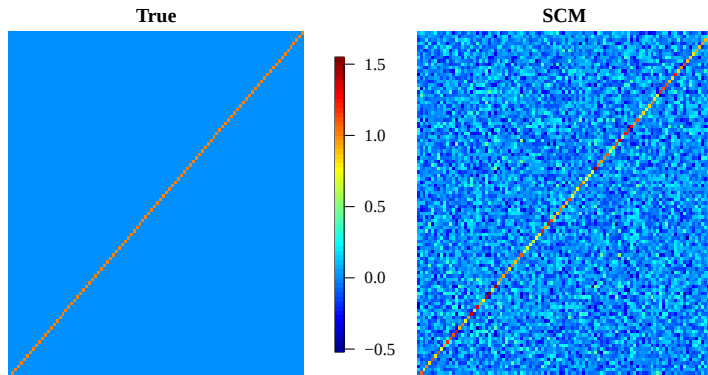
so  $\hat{\Sigma}$  is **singular**.

This is not just a numerical problem: there is simply not enough information in the data to estimate an unrestricted  $p \times p$  covariance matrix reliably.

# SCM Breakdown in High Dimensions

As  $p$  grows relative to  $n$ :

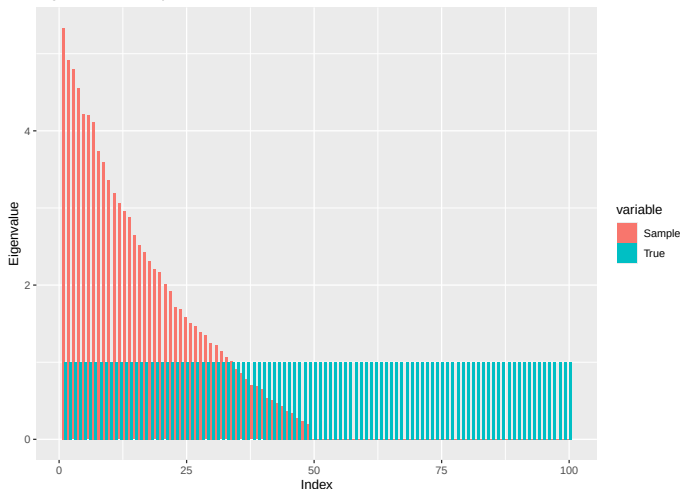
- ▶  $\hat{\Sigma}$  can be **noisy and unstable**, even when  $p < n$
- ▶ Eigenvalues and  $\hat{\Sigma}^{-1}$  can be poorly estimated
- ▶ When  $p \geq n$ ,  $\hat{\Sigma}$  becomes singular



# SCM Breakdown: Eigenvalue Distortion

Even in the simple case  $\Sigma = I$ , when  $n = 50 < p = 100$ , the sample eigenvalues are **severely distorted**: many are zero, while the nonzero eigenvalues are widely dispersed around the true value of 1.

Eigenvalues: Sample vs. True



# Why Are Sample Eigenvalues Unstable?

Even when the true covariance matrix is

$$\Sigma = I,$$

the eigenvalues of  $\hat{\Sigma}$  can be widely dispersed when  $p$  is large relative to  $n$ .

When

$$p, n \rightarrow \infty, \quad \frac{p}{n} \rightarrow c,$$

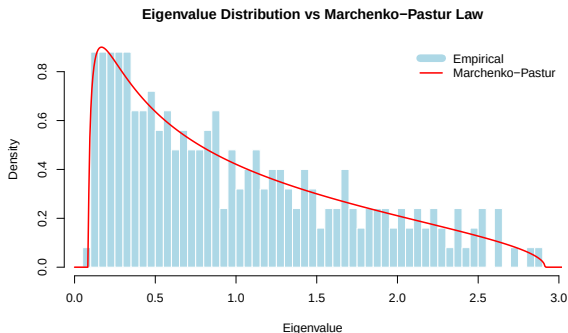
random matrix theory shows that the empirical eigenvalues follow the [Marchenko–Pastur law](#).

**Practical message:** large and small sample eigenvalues may reflect [sampling noise](#), not genuine structure.

# Marchenko–Pastur Law

For  $\Sigma = I$  and  $p/n \rightarrow c \in (0, 1)$ , the sample eigenvalues spread over

$$(1 - \sqrt{c})^2 \leq \hat{\lambda} \leq (1 + \sqrt{c})^2$$



Larger  $p/n \Rightarrow$  greater eigenvalue dispersion.

# Shrinkage Estimation: Stabilizing the SCM

A simple strategy is to combine the sample covariance matrix with a more stable target:

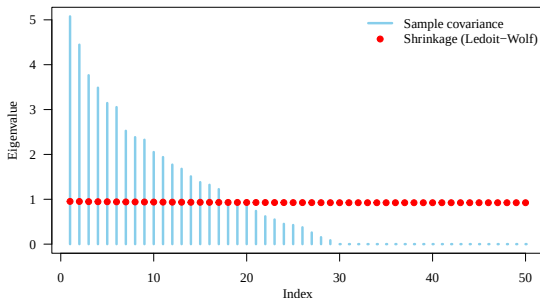
$$\hat{\Sigma}_{\text{shrink}} = (1 - \lambda)\hat{\Sigma} + \lambda \mathbf{T}, \quad 0 \leq \lambda \leq 1.$$

- ▶  $\mathbf{T}$ : stable target, such as the identity matrix
- ▶  $\lambda = 0$ : use the SCM completely
- ▶  $\lambda = 1$ : use the target completely
- ▶ Intermediate  $\lambda$ : trade some bias for lower variance

**Main idea:** a slightly biased estimator can have substantially better prediction and numerical stability than the SCM.

# Ledoit–Wolf Shrinkage

Ledoit–Wolf chooses the shrinkage intensity  $\lambda$  from the data to balance bias and variance.



- Simulation:  $p = 50$ ,  $n = 30$
- Shrinkage stabilizes eigenvalues and improves conditioning

# Sparse Precision Matrices

The inverse covariance matrix

$$\Theta = \Sigma^{-1}$$

is called the **precision matrix**.

For multivariate normal data,

$$\Theta_{ij} = 0 \iff X_i \perp X_j \mid \text{all other variables.}$$

Thus, zeros in  $\Theta$  describe **conditional independence** relationships.

A sparse precision matrix provides both **regularization** and an interpretable network of relationships among variables.

# Graphical Lasso

The graphical lasso estimates a sparse precision matrix by solving

$$\hat{\Theta} = \arg \max_{\Theta \succ 0} \left\{ \log \det(\Theta) - \text{tr}(\hat{\Sigma}\Theta) - \rho \|\Theta\|_1 \right\}.$$

- ▶ First two terms: fit the multivariate normal model
- ▶  $\rho \|\Theta\|_1$ : penalizes complexity
- ▶ Larger  $\rho$ : more entries estimated as zero
- ▶ Zeros correspond to missing edges in the conditional independence graph

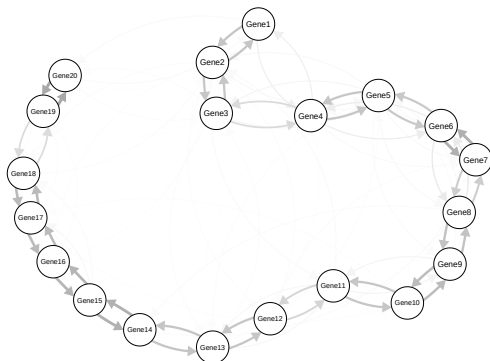
This is the covariance analogue of the broader [regularization and sparsity](#) ideas used throughout data science.

## Graphical Lasso Application

Consider data with an AR(1) covariance structure:  $\Sigma_{ij} = \rho^{|i-j|}$ .

- ▶ Although  $\Sigma$  is dense, its precision matrix  $\Theta = \Sigma^{-1}$  is **sparse**
- ▶ Graphical Lasso estimates  $\Theta$  while encouraging many off-diagonal entries to be zero
- ▶  $\hat{\Theta}_{ij} \neq 0 \Rightarrow$  variables  $i$  and  $j$  are conditionally dependent

Graphical Lasso Network (nonzeros in precision)



# Summary

- ▶ The sample covariance matrix (SCM) is the classical estimator, with properties governed by the Wishart distribution
- ▶ Inference tools include confidence intervals, hypothesis tests, and bootstrap methods
- ▶ In high-dimensional settings, the SCM becomes unstable or singular, motivating regularization
- ▶ Shrinkage estimators (e.g., Ledoit-Wolf) improve conditioning; sparse methods (e.g., graphical lasso) enable interpretable structure learning
- ▶ Practical implementations are available in R: `cov`, `cov.shrink`, `glasso`, `boot`